

FCM-RT: Real-Time Feature Coding for Machines

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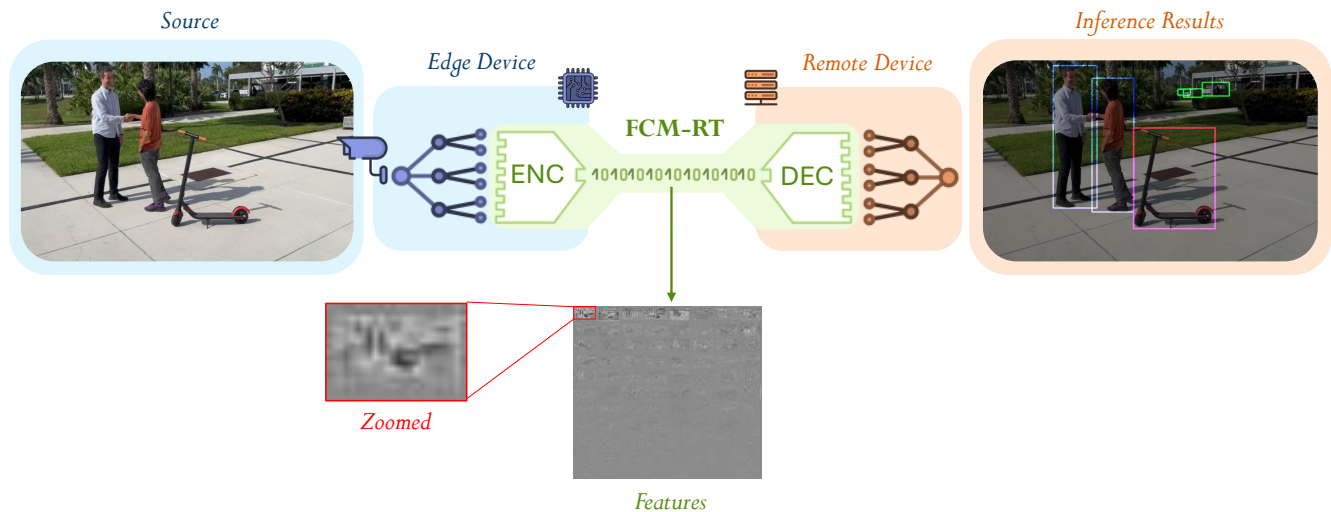


Figure 1: Diagram of the split inference implementation using FCM-RT.

Abstract

As numerous edge devices start implementing intelligent components, the challenges of energy consumption, bandwidth efficiency, and privacy gain significance. One proposed solution relies on the paradigm of split inference, which optimizes the delegation of the computational load between edge and remote devices. We developed and implemented the standard-compliant split inference system with an encoder and decoder capable of real-time streaming and processing. Our system outperforms state-of-the-art video compression implementations by an average of 83% bitrate reduction, while preserving privacy. We demonstrate the system's real-time performance on consumer devices, with interactive visualizations of object detection and segmentation, incorporating real-time metrics. Demo video: https://youtu.be/bmCbUo_ZWWU

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CCS Concepts

• Computing methodologies → Computer vision; Image compression.

Keywords

Feature coding, low runtime, live streaming, split inference, collaborative inference, coding for machines, FCM, MPEG-AI

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1 Introduction

In recent years we have been witnessing the emergence of large numbers of edge devices that are capable of acquiring and processing visual signals. Devices such as smart cameras, drones, and robots need to process visual information efficiently, considering energy and bandwidth limitations. Traditional video coding is tailored towards human consumption and is not optimized for machine use cases [2].

One promising solution is the implementation of the split inference paradigm. In a classical system that can be called remote inference, an edge device acquires, encodes and sends the video bitstream to a remote device. The remote device decodes the bitstream and passes the decoded video through the task neural network (inference). With the increasing number of edge devices, the computational burden at the remote increases significantly. In addition, new computing architectures that are available on the edge devices (such as Neural Processing Units, NPUs) remain underutilized in the remote inference systems. Split inference systems delegate some of the inference process to the edge device (hence, “splitting” the task neural network) as depicted in Figure 1.

Moving Picture Experts Group (MPEG) initiated the Feature Coding for Machines (FCM) standardization [4]. The standardization activity focuses on defining normative bitstream and compression tools specifically for neural intermediate features in machine vision systems. The standardization began in 2022, with the International Standard expected in 2027. As part of the standardization process, the reference software model called Feature Coding Test Model (FCTM) [4] was implemented as part of a framework for evaluating proposed methods.

In the timeline of most international standards, a significant milestone is the successful development of normative software implementation that supports critical use cases. One of the recent examples is the state-of-the-art video coding standard Versatile Video Coding (VVC), which was finalized in 2020, but the efficient codec implementation was not completed until more than a year later [8]. Real-time coding is a critical requirement for many use cases. Since the FCM use cases include many scenarios in which real-time streaming is critical for task completion, it is obvious that the success of the standard heavily relies on the normative implementation that supports real-time processing.

To address this, we designed and developed FCM Real-Time (FCM-RT). This is a real-time implementation of the FCM standard, which well surpasses requirements for real-time encoding and decoding, even on Full HD video and consumer devices. Its low runtime makes it immediately ready for live streaming, which we demonstrate with an end-to-end split inference system. With FCM-RT, users may interact with a live FCM stream to understand the fundamental shifts between traditional (local and remote) inference and split inference.

2 System Framework

FCM-RT is based on the reference software FCTM version 6.1 [4], which integrates trained neural networks and a video-based inner codec. The neural components use CUDA acceleration [5], making the GPU/NPU architecture desirable for implementation. FCTM requires processing of the entire video sequence before encoding, preventing low delay encoding. To overcome this limitation, our system processes video at frame-level, enabling encoding and decoding without future references. The optimized design is implemented in a way that keeps the codec standard-compliant. Additionally, the inner encoder in FCTM, VTM, is replaced with x265 [7], utilizing HEVC in place of VVC to enable real-time processing.

To demonstrate the capabilities of the proposed codec, a live system was developed for object detection using Faster R-CNN [6]

Table 1: Bitrate (kbps) of Codecs at Detection Performance Levels

mAP	FCM-RT (Ours)	VVC	HEVC
65%	-	-	-
60%	216	1478 (5.8×)	4217 (18.5×)
55%	103	743 (6.2×)	2111 (19.5×)
50%	67	450 (5.7×)	1546 (22.1×)
45%	48	317 (5.6×)	1212 (24.3×)
40%	36	235 (5.5×)	937 (25.0×)

and object segmentation using Mask R-CNN [3], on two machines connected over a local network. The first machine is the edge device, equipped with a laptop-grade NVIDIA 4060 GPU and a live-recording camera. It performs the initial stage of neural processing by applying the ResNeXt-101 backbone to the video feed and encodes the processed frames. The FCM bitstream is generated in chunks and immediately streamed to the second machine. This second machine is the remote device, equipped with a laptop-grade NVIDIA 3070 Ti GPU. It decodes the stream and completes the remaining inference to produce detection or segmentation results.

3 System Demonstration

When the system camera captures a live video feed, users may interact with the output by viewing inference results (rendered as detected objects). The system also allows toggling between object detection and segmentation tasks. Additionally, it provides access to the real-time feature stream at an intermediate encoder stage that represents the transmitted data. This stream demonstrates that the features preserve user privacy while retaining semantic information sufficient for machine vision. The system also reports bitrate and frames-per-second (FPS) metrics to illustrate performance.

4 Experimental Results

Extensive testing is conducted to evaluate the suitability of the proposed codec implementation. A key objective is bitrate saving compared to remote inference. Benchmarks are performed against state-of-the-art low-delay encoders for HEVC and VVC, specifically x265 and VVenC [8], respectively.

Object detection using Faster R-CNN is selected as the evaluation task, with mean average precision (mAP) used as the performance metric. The dataset used is SFU [1], which consists of annotated uncompressed videos. For FCM-RT, which requires network partitioning, the split is applied after the ResNeXt-101 feature backbone, where bitrate is varied through quantization level.

A dense set of target bitrates is tested to assess the rate-distortion tradeoff. Table 1 summarizes the average bitrate required by each codec across varying detection performance levels, with approximately 65% mAP representing uncompressed performance. The proposed system achieves 5.5–6.2× lower bitrate than VVC and 18.5–25.0× lower bitrate than HEVC. Encoder and decoder frame rates are also measured: for 1080p (Full HD) resolution, the system achieves 72 FPS encoding and 33 FPS decoding.

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